

Grade 8
Unit 7
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

Gabriella is paying off her car loan. The table of values represents the amount owed, y , and the amount of time, x , in months. Use the table of values to answer questions 1–7.

Months	Owed
2	\$14,500
6	\$13,500
8	\$13,000
12	\$12,000

1. What is the slope of this relationship?

$$\frac{-1000}{4} = -250$$

2. What is the y -intercept of the relationship?

(0,15,000)

Use the graph to match each key feature with its interpretation.

Key Feature	Interpretation
3. x -intercept	A. Gabriella bought the car for \$15,000.
4. y -intercept	B. Gabriella will pay off the loan in 5 years.
5. slope	C. The amount owed is decreasing.
6. sign of the slope	D. Gabriella is paying \$250 per month.

7. Explain why negative values for x and y do not make sense in this context.

Negative y doesn't make sense because she can't owe negative money. Negative x doesn't exist because time isn't negative. (Answers may vary)

Bald cypress trees were planted along the riverbanks to combat erosion. The table of values depicts, the number of years since the tree was planted, x and the height, y , in feet.

Years	Height (ft)
4	8
12	20
15	24.5
20	32

8. Determine the slope of this relationship and interpret its meaning in the given context.

$$\frac{20-8}{12-4} = \frac{12}{8} = \frac{3}{2} \quad \text{The tree grows 1.5 ft per year.}$$

9. Explain why the x -intercept would not make sense in the context of this problem.

It didn't start or reach a height of 0.

10. Jack and Diane are racing to the finish line. Based on the information in the table, who will win the 10-mile race?

Jack's Race	Diane's Race										
<table> <tr> <th>Minutes</th><th>Mile</th></tr> <tr> <td>20</td><td>3</td></tr> <tr> <td>40</td><td>6</td></tr> <tr> <td>60</td><td>9</td></tr> <tr> <td>80</td><td>12</td></tr> </table>	Minutes	Mile	20	3	40	6	60	9	80	12	Diane ran the entire race at a speed of 8 miles per hour.
Minutes	Mile										
20	3										
40	6										
60	9										
80	12										

9 miles per hour

Jack is faster

$$\frac{6-3}{40-20} = \frac{3}{20}$$

$$\frac{3 \text{ miles}}{20 \text{ minutes}} = \frac{x}{60 \text{ minutes}}$$

$$x = 9$$